

A Hot Bath: Conserving Water Without Losing Uniformity

Summary

A warm mean does not ensure a uniformly warm bath. We minimize replacement water subject to a lower limit in every cell, and upper-temperature and spread limits outside a 0.15 m inlet zone. Stored heat favors waiting; finite transport can make distant water cold before that waiting period ends.

We prove a coast-then-hold policy optimal for a well-mixed bath, then use a three-dimensional finite-volume thermal network with coefficients derived from textbook correlations and tied to a published immersion study where one exists (Section 2). The best constant rate is compared with an optimized piecewise-constant schedule; neither is proved optimal among all controls.

In a 164.25 L water-volume scenario lasting 30 minutes, with an initial temperature of 40°C and a 39°C lower limit, the best accepted constant-rate candidate adds **24.14 L** at 0.805 L/min from the start, and a 6-segment schedule with a 0.1°C design margin needs **21.48 L**, 11% less. The well-mixed optimum is 16.01 L, while an independent energy argument gives a 15.41 L lower bound for the spatial problem. A finer mesh changes the constant-rate result by 0.11%, and the selected scheduled policy passes independent continuous-time bounds on all three tested meshes.

Across 64 Sobol draws over the stated parameter ranges, 41 yield an accepted constant-rate candidate, needing 12–31 L (5th–95th percentile). Weak mixing often defeats this finite search. Motion improves transport, but the associated extra surface loss can offset the saving. The assumed foam layer reduces constant-flow demand to 10.4 L; this is a scenario, not a measured additive effect.

A distinct axial-dispersion model confirms remote cooling and changes the preferred start from immediate supply to a ten-minute wait as mixing strengthens. Its overflow is cooler than the mean, as is the adopted three-dimensional baseline: the spatial water penalty cannot simply be attributed to hot-water short circuit. Cooling alone cannot identify the inlet route. At common reserves, passive/pulse candidates use 26.09/23.63 L; a reusable 6 L pulse pays from unchanged use 3 with equal resets. Probe updates reduce the passive backup to 23.59/23.16 L in two nominal noise cases. Two fixed realized schedules have narrow continuous-parameter bounds; changed physics can invalidate certificate reuse.

Keywords: thermal network; energy balance; lower bound; optimal control; sensitivity analysis

Contents

1	Define the decision before optimizing	3
2	Physical assumptions and scenario inputs	4
2.1	Where the coefficients come from.	5
3	A transparent well-mixed benchmark	5
4	What can be proved about water use?	6
5	A conservative model in three dimensions.	7
5.1	Spatial evidence: the mean is not the whole bath	8
6	Inlet transport, solver and strategy search	9
7	The spatial result and the mixing penalty	9
7.1	A time-varying schedule uses less water.	10
7.2	What does staying close to the start temperature cost?.	12
7.3	Can a user follow the schedule?	13
7.4	Measurement ambiguity changes the decision	13
8	Separate geometry, size and body effects.	15
9	Motion and a bubble layer can change the policy	16
10	Validation: independent evidence and remaining error	17
11	Resolution, uncertainty and transfer to a real bath	17
11.1	Carry structural ambiguity into the decision	18
11.2	Use observations to change the next action	19
11.3	Check transfer beyond the parameter grid.	20
11.4	Challenge the transport mechanism.	21
12	Conclusions and a policy with clear scope.	22
13	A warmer bath, with less replacement water.	23
14	References and reproducible algorithm	24
	Appendix A. Scenario definitions.	25
	Report on Use of AI	26

1 Define the decision before optimizing

The task is to preserve both warmth and spatial uniformity in an overflowing, unheated tub, and to examine geometry, the bather and motion, and a bubble-bath layer [1]. The report separates physical requirements from preference assumptions. There is no supplied temperature record or measured heat-transfer coefficient to fit.

The decision variables are an inlet flow rate and the time at which a constant trickle begins. The tub is already full: added water displaces an equal volume through the overflow. The horizon is 1,800 s. Our baseline requires every cell average to stay at or above 39°C, and requires the cell averages outside a 0.15 m jet-mixing zone around the inlet to stay at or below 41°C and within an instantaneous spread of 1.5°C. These choices operationalize comfort; they are not medical limits or numbers specified by the problem, and the jet zone is an assumption justified in Section 2.1.

$$J = 1000 \int_0^{t_f} q(t) dt \quad (1)$$

$$T_i(t) \geq T_{\min} \quad (\forall i), \quad T_i(t) \leq T_{\max}, \quad \max_{i \in \Omega} T_i(t) - \min_{i \in \Omega} T_i(t) \leq \Delta \quad (i \in \Omega) \quad (2)$$

Here Ω is the set of cells outside the inlet jet zone, q is in m³/s and J is in litres. We prioritize the least water within the constraints, rather than assigning arbitrary weights to unlike units. Tightening the temperature tolerance is a separate scenario. Zero flow is admitted: if the initial stored heat suffices, using no added water is globally water-minimal.

Table 1 maps each requirement to its answer.

Table 1. Where each requirement is answered

Requirement	Where answered
Temperature in space and time	Sections 3, 5–7; Figures 1–2
Water-use strategy and its scope	Sections 4, 6–7 and 12
Tub/body geometry, size and temperature	Section 8
Motion and bubble-bath additive	Section 9
Validation and sensitivity	Sections 8–11
One-page non-technical explanation	Section 13

Spatial temperature means a control-volume average. It does not bound the unresolved temperature of a faucet jet or a skin-contact film. This distinction determines which practical conclusions the simulation can support.

Modeling options. The lumped model supplies a proved benchmark; the conservative network resolves cold regions with an explicit mixing parameter [5]. A velocity solver needs jet and turbulence data unavailable here. Table 2 compares their roles.

Table 2 contrasts spatial resolution and data requirements.

Table 2. Modeling options compared

Option	Resolves	Evidence it needs	Role here
Lumped (Newton)	Mean temperature	Two loss coefficients	Proved benchmark
Thermal network	Cells, heat paths, overflow	Loss, body and mixing coefficients	Decision model
Flow solver	Velocity and buoyancy	Turbulence closure, jet data	Not used

2 Physical assumptions and scenario inputs

The problem supplies no calibrated coefficients. Section 2.1 separates external anchors from assumptions; scenarios test the important ones.

Table 3 separates assumed inputs from anchored coefficients.

Table 3. Baseline inputs

Quantity	Baseline / units	Status
Tub $L \times W \times H$	$1.50 \times 0.65 \times 0.23$ m	Assumed geometry
Displaced body volume / area	$0.060 \text{ m}^3 / 1.15 \text{ m}^2$	Assumed
Water density / heat capacity	$1000 \text{ kg/m}^3 / 4180 \text{ J/(kg K)}$	Rounded constants
Air / skin / inlet temperature	$22 / 34 / 50^\circ\text{C}$	Fixed reservoirs
Surface / wall / body coefficient	$25 / 6.5 / 25 \text{ W/(m}^2 \text{ K)}$	Correlation-based (next page)
Mixing diffusivity D	$0.001 \text{ m}^2/\text{s}$	Uncalibrated closure; range tested
Initial / lower / upper limit	$40 / 39 / 41^\circ\text{C}$	Preference scenario
Time / allowed spread	$1800 \text{ s} / 1.5^\circ\text{C}$	Preference scenario

A1. Constant density and heat capacity. Water properties change by under 1% between 39 and 41°C , and the constants are rounded values, not an evaluated IAPWS table [3]. *Reason:* the resulting capacity error is small next to the loss coefficients.

A2. Linear exchange. Newton's law [2] approximates the narrow temperature span. Surface loss includes convection, radiation and evaporation once; evaporative volume change is neglected. Room-temperature linearization understates near- 40°C sensitivity by about 7% of total loss in an independent estimate, within the tested coefficient range.

A3. Skin held at a fixed temperature. *Reason:* thermoregulation is outside the problem; skin temperature is varied (32 and 36°C) instead of resolved.

A4. Prescribed surface stream. Water traverses the top row nearest the centre line (half a cell off it for even row counts). Alternative paths and whole-section flow are tested in Section 11; the route is assumed, not measured.

A5. Mixing as one effective diffusivity D . Motion enters through D , with a separate surface-

loss scenario. *Reason:* molecular diffusion alone would not represent circulation; D is anchored only by an order-of-magnitude scaling (Section 2.1) and is tested over a range; vertical mixing is tested separately in Section 9.

2.1 Where the coefficients come from

No temperature record exists to fit, so each coefficient is derived from a textbook relation or tied to a published measurement where one exists, and given a range; where nothing exists (mixing, the exposed fraction, the air and skin temperatures) the value is an assumption and is labelled as one. Surface loss sums natural convection above a hot horizontal surface ($Nu = 0.15 Ra^{1/3}$ for $10^7 < Ra < 10^{11}$, length A/P [5]; here $Ra = 1.9 \times 10^7$), linearized radiation (emissivity 0.96) and evaporation by the Lewis analogy. For open water at 40°C in 22°C air at 50% humidity the parts are 4.7, 6.1 and 24.6 W/(m² K) (0.66 kg/m² h evaporated), 35.4 in total. A bather covers part of the surface; with an assumed exposed fraction of 0.7 the central value is 24.8, and we use 25 within the range 17–37, which spans exposed fractions of 0.5 to 1.0. Evaporation correlations differ by tens of percent, so this is a scenario range, not a measured one.

A 5 mm shell (0.19 W/(m K)), with inside/outside films 300/8 W/(m² K), gives a series coefficient 6.5, with range 4.7–8.4; these are typical inputs. Menzies et al. observed rectal rise $0.9 \pm 0.3^\circ\text{C}$ after 30 min shoulder-depth immersion at 40°C [6]. For mass 73 kg and heat capacity 3470 J/(kg K), an assumed mean rise 1–2°C implies uptake 141–281 W and equivalent exchange 20–41 W/(m² K) at fixed 34°C skin. We use 25. Mean rise and driving difference are assumptions: this is a scale anchor, not calibration. Body capacity is about 37% of water capacity; its delayed warming is tested in Section 11.1.

The assumed scaling $D \approx 0.1u'\ell$ gives $1 \times 10^{-4}/1 \times 10^{-3}/6 \times 10^{-3}$ m²/s for velocity/eddy-length pairs (0.02 m/s, 0.05 m), (0.1, 0.1) and (0.3, 0.2). We use 0.001 and test 0.0003–0.003; these are circulation scenarios, not measured diffusivities.

The 39–41°C window is a preference, not a safety standard [6]. Inlet-cell maxima increase 40.55/40.76/41.00°C on the three meshes: the point source heats less water per cell. Ceiling/spread therefore exclude a 0.15 m inlet zone, assuming no bather occupies the jet; the floor covers every cell, as required by Section 4. Section 11 checks constrained quantities.

Table 4 summarizes the coefficient anchors and scenario ranges.

3 A transparent well-mixed benchmark

Table 5 defines storage, transport and exchange symbols.

Let C be total water heat capacity, H_a the air/shell conductance and H_b the body conductance. A well-mixed overflow has the same temperature as the bath. Integrating the physical energy balance gives

$$C\dot{T} = H_a(T_a - T) + H_b(T_b - T) + \rho c_p q(T_{in} - T) \quad (3)$$

Table 4. Coefficient anchors and the values used

Coefficient	Relation or anchor	Derived value	Used (range tested)
Surface, W/(m ² K)	Convection + radiation + evaporation [5]	35.4 open; 24.8 at 70%	25 (17–37)
Shell, W/(m ² K)	Series resistance, typical shell values	6.5 (4.7–8.4)	6.5 (4.5–8.5)
Body, W/(m ² K)	Uptake implied by core rise [6]	20–41 at fixed skin	25 (12–40)
Mixing D , m ² /s	Mixing-length scaling	1×10^{-4} to 6×10^{-3}	0.001 (0.0003–0.003)

Table 5. Symbols

Symbol	Meaning
C	Heat capacity of the water, J/K
H_a, H_b	Air/shell and body conductance, W/K
T_a, T_b, T_{in}	Room, skin and inlet temperature, °C
T_{min}, T_{max}	Lower and upper limit, °C
q, J	Inlet flow, m ³ /s; added water, L
D	Effective mixing diffusivity, m ² /s

$$C = \rho c_p (LWH - V_b), \quad H_b = h_b A_b \quad (4)$$

$$H_a = h_s f LW + h_w \{LW + 2H(L + W)\} \quad (5)$$

The foam multiplier f equals one without a layer. Setting $q = 0$, define $H = H_a + H_b$ and $T_e = (H_a T_a + H_b T_b)/H$. The cooling solution and first time to reach the lower limit are

$$T(t) = T_e + (T_0 - T_e)e^{-Ht/C} \quad (6)$$

$$t_c = \frac{C}{H} \log \frac{T_0 - T_e}{T_{min} - T_e} \quad (7)$$

Baseline conductances are $H_a = 37.14$ W/K and $H_b = 28.75$ W/K, and $C = 686,565$ J/K. Thus $T_e = 27.24^\circ\text{C}$ and $t_c = 14.17$ min.

A heat-loss model must approach an environmental equilibrium rather than zero Celsius. This analytic boundary also supplies a direct check on signs and units. The benchmark describes mixing perfectly; the spatial model will test the cost of departing from that assumption.

4 What can be proved about water use?

Let $\ell(T) = H_a(T - T_a) + H_b(T - T_b)$ be the heat loss of the mixed bath, and let the inlet be hotter than every temperature considered. **Proposition 1.** Coasting to the lower limit and then holding it uses the least water, provided the holding rate is within the faucet bound. The proof has four steps.

Proof. Step 1, an identity. Dividing the balance by $T_{\text{in}} - T$ and integrating over the horizon gives

$$\int q \, dt = \frac{C}{\rho c_p} \log \frac{T_{\text{in}} - T_0}{T_{\text{in}} - T_f} + \int \frac{\ell(T)}{\rho c_p (T_{\text{in}} - T)} \, dt \quad (8)$$

Step 2, monotonicity. The integrand $\ell(T)/(T_{\text{in}} - T)$ has derivative $[H_a(T_{\text{in}} - T_a) + H_b(T_{\text{in}} - T_b)]/(T_{\text{in}} - T)^2 > 0$, so it increases with T , and the logarithmic term increases with the final temperature. Water use is therefore an increasing functional of the temperature path.

Step 3, the lowest admissible path. Because $q \geq 0$, comparison with the no-flow solution gives $T(t) \geq T_c(t)$, the cooling curve of Section 3, and feasibility requires $T(t) \geq T_{\text{min}}$. The pointwise lowest admissible path is $\max(T_c(t), T_{\text{min}})$: coast until T_c reaches T_{min} , then hold.

Step 4, attainability. Holding $T = T_{\text{min}}$ means $\dot{T} = 0$, which needs the rate

$$q_{\text{hold}} = \frac{\ell(T_{\text{min}})}{\rho c_p (T_{\text{in}} - T_{\text{min}})} \quad (9)$$

When this rate is within the bound, the path is feasible and, by Step 2, optimal.

□

For the baseline, the mixed-model optimum adds 16.01 L: wait 14.17 min, then supply 1.012 L/min. Optimality is established for the mixed model; spatial policies are assessed separately.

Proposition 2 (energy lower bound). In the spatial model, suppose every cell satisfies $T_i(t) \geq T_{\text{min}}$. Summing the cell balances gives $\sum_i C_i T_i(t_f) - C T_0 = - \int L \, dt + \rho c_p \int q (T_{\text{in}} - T_{\text{out}}) \, dt$, with L the total loss. Each loss term increases with its cell temperature, and H_a and H_b are the sums of the cell conductances, so $L(t) \geq \ell(T_{\text{min}})$. The outlet cell satisfies $T_{\text{out}} \geq T_{\text{min}}$, so $T_{\text{in}} - T_{\text{out}} \leq T_{\text{in}} - T_{\text{min}}$, and the left side is at least $C(T_{\text{min}} - T_0)$. Combining these three facts,

$$J \geq \max \left(0, \frac{1000[\ell(T_{\text{min}})t_f - C(T_0 - T_{\text{min}})]}{\rho c_p (T_{\text{in}} - T_{\text{min}})} \right) \quad (10)$$

The resulting 15.41 L is a genuine conditional lower bound, valid for any control, but it is not tight: the initial water is hotter than the floor and loses more heat, and limited mixing adds further cost. Section 7.1 shows how much of the gap an optimized schedule recovers.

5 A conservative model in three dimensions

The rectangular water envelope is divided into $8 \times 4 \times 3$ cells. Each has a capacity C_i and exchanges heat only across common faces. The cells whose centres lie within 0.15 m of the inlet cell form the jet zone (2, 9 and 22 cells on the $8 \times 4 \times 3$, $12 \times 6 \times 4$ and $16 \times 8 \times 6$ meshes); Ω is the rest. Top cells lose heat to the room; bottom and side faces lose heat through the shell. The bather is represented by a smooth, three-dimensional displacement field and a distributed skin contact term.

$$C_i \dot{T}_i = \sum_j g_{ij} (T_j - T_i) + H_{a,i} (T_a - T_i) + H_{b,i} (T_b - T_i) + S_i \quad (11)$$

$$g_{ij} = \rho c_p D \frac{A_{ij}}{d_{ij}} \min(\phi_i, \phi_j), \quad C_i = \rho c_p V_i \quad (12)$$

Here ϕ is fluid fraction, A_{ij} a face area and d_{ij} the center separation. Symmetric g_{ij} guarantees that internal heat exchange cancels when the equations are summed. D is an effective mixing closure; molecular diffusion alone would not represent motion or buoyant circulation.

$$w_i \propto \exp \left[-\frac{1}{2} \sum_{k=1}^3 \left(\frac{x_{ik} - b_k}{s_k} \right)^2 \right], \quad \sum_i w_i = 1 \quad (13)$$

$$V_i = V_{\text{cell}} - V_b w_i, \quad H_{b,i} = h_b A_b w_i \quad (14)$$

Baseline $\mathbf{b} = (0.55L, W/2, 0.45H)$, with widths $\mathbf{s} = (0.40, 0.16, 0.12)$ m. Fluid fraction is $\phi_i = V_i/V_{\text{cell}}$. The center $\mathbf{b}$ and widths $\mathbf{s}$ set the spatial distribution of body effects. Body volume and contact area remain independent inputs: shape is varied through $\mathbf{s}$ at fixed volume and area. This is a homogenized immersed-body representation, not an anatomically resolved obstruction. A cell whose occupied fraction reaches 0.95 is rejected rather than assigned a negative capacity.

A fixed envelope volume is preserved by balancing inlet and overflow. In the baseline the exact displaced-water volume is 164.25 L. Shape scenarios preserve envelope volume when isolating aspect-ratio effects. Volume scenarios deliberately change water depth and hence both storage and side-wall area.

5.1 Spatial evidence: the mean is not the whole bath

Figure 1 locates cold regions across layers on one scale.

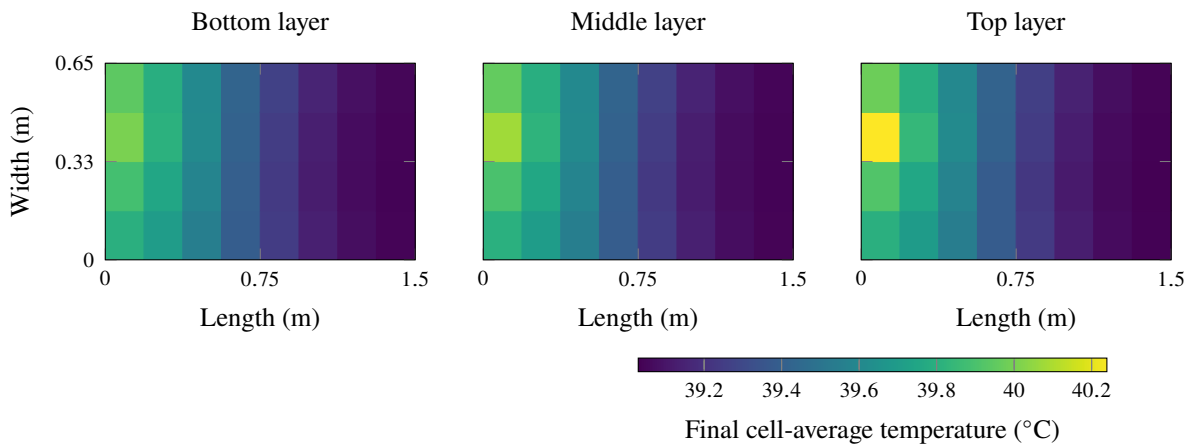


Figure 1. Final temperatures in the three horizontal cell layers. All layers use one color scale; the top layer contains the prescribed inlet-to-overflow stream. Geometry is in metres, and values are cell averages.

The heat map and Figure 2 use the same archived trajectory. Exchange couples the horizontal layers into one three-dimensional network. The cold-region locations explain information lost by a mean or range; the mesh table tests resolution. These are homogenized cell averages, not resolved anatomy, recirculation or a faucet jet.

6 Inlet transport, solver and strategy search

Inlet and overflow are on opposite ends of a surface stream. Every path edge carries the same q : the first cell receives hot water, interior cells receive upstream water and lose the same volume downstream, and the last cell discharges through the overflow.

$$S_1 = \rho c_p q (T_{in} - T_1), \quad S_i = \rho c_p q (T_{i-1} - T_i) \quad (15)$$

$$\sum_i C_i \dot{T}_i = - \sum_i H_{a,i} (T_i - T_a) - \sum_i H_{b,i} (T_i - T_b) + \rho c_p q (T_{in} - T_{out}) \quad (16)$$

This prescribed stream permits short circuit but does not establish excess outlet heat loss. Section 11.4 diagnoses its sign against a distinct whole-section closure; Section 11.1 changes the internal path at fixed inlet and overflow. Neither route is an inferred velocity field.

For a fixed flow, the network is affine linear. Augmenting the state by a constant one permits matrix-exponential propagation, avoiding a large forward-Euler step restriction [4]. A delayed-start policy has two exact constant-control segments.

$$\dot{\mathbf{T}} = A(q)\mathbf{T} + \mathbf{b}(q), \quad \mathbf{z}(t + \tau) = e^{\tilde{A}(q)\tau} \mathbf{z}(t) \quad (17)$$

Starts are tested at 0, 120, ..., 1200 seconds. For each, rates from 0 to 3 L/min are bracketed on a 0.2 L/min grid; the first crossing of the lower-temperature requirement is refined by Brent's root method. The search targets $T_{\min} + 0.03^\circ\text{C}$ as numerical reserve, then checks the upper-temperature and spread limits at 5-second output intervals. Starts that already violate the floor are rejected.

This is a reproducible candidate search. Nonmonotone flow responses, untested delays, pulsed inputs, inlet relocation and feedback are not excluded by the calculation. A failed search means no candidate was accepted, not that every possible action is infeasible. The independent checks further examine the chosen trajectory between samples.

7 The spatial result and the mixing penalty

Figure 2 contrasts the mean, coldest cell and spatial range.

Table 6 reports the constant-rate trajectory and water use.

Table 6. Baseline result under the best constant rate

Result	Baseline
Rate / start	0.805 L/min / 0.0 min
Added water	24.14 L
Minimum / maximum	39.030 / 40.318°C
Maximum simultaneous spread	0.967°C
Final volume-weighted mean	39.418°C

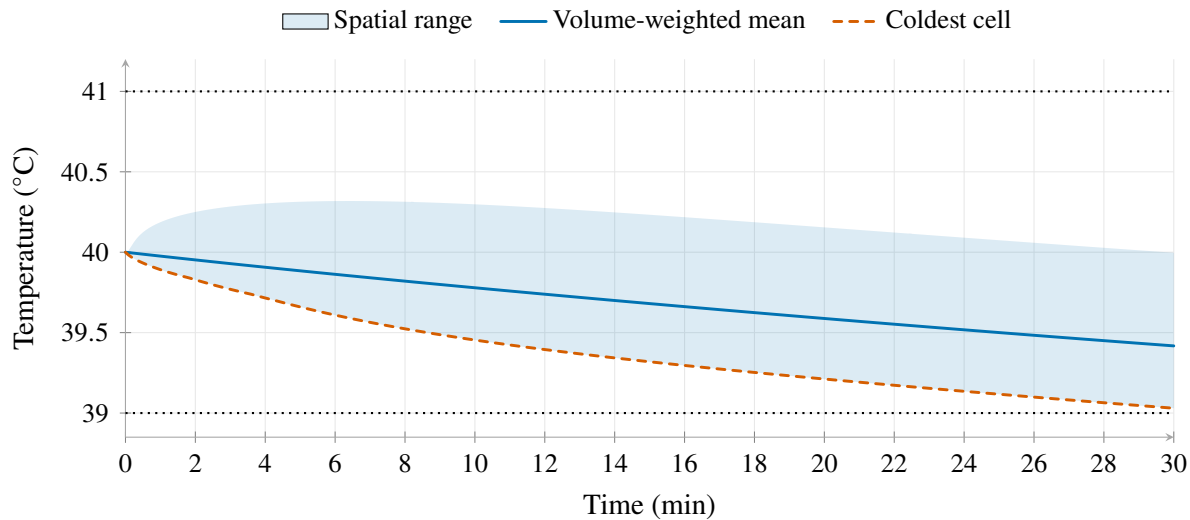


Figure 2. Matrix-exponential cell temperatures under the selected policy. The shaded range covers the coldest cell and the hottest cell outside the inlet jet zone; the mean alone would hide cold and hot locations.

Within the constant-rate family, starting the trickle immediately ranks ahead of delayed starts here, even though waiting is optimal in the ideal mixed model. The distant water needs time to receive heat; larger late rates create a warmer inlet region before they solve the cold-region constraint. This is a concrete consequence of resolving space.

Holding a perfectly mixed bath at 40°C throughout would use 36.22 L. This is a stricter reference service, not a like-for-like optimum. Our accepted 1°C cooling allowance uses less water partly because it provides a different service. The comparable mixed model with the same lower limit uses 16.01 L. The gap also reflects our restricted control family and 0.03°C numerical reserve; it cannot be attributed purely to imperfect mixing.

7.1 A time-varying schedule uses less water

Constant-rate policies are a narrow family. We therefore optimize piecewise-constant inlet flow in K equal segments ($K = 3, 6, 12$) by sequential quadratic programming, minimizing added water under the same floor and outside-jet limits (with the 0.03°C floor reserve), sampled every 5 s, from four starting schedules. The problem is non-convex: each result is the best local optimum found, with no global claim.

Table 7 separates candidates from proved bounds.

Table 7. Constant rate, scheduled flow and the bounds

Policy	Water (L)	Change	Min / max (°C)	Spread (°C)
Best constant rate	24.14	—	39.03 / 40.32	0.97
3 segments, coarse candidate	19.75	-18.2%	39.03 / 40.74	1.50
6 segments, coarse candidate	19.50	-19.2%	39.03 / 40.73	1.50
12 segments, coarse candidate	19.35	-19.9%	39.03 / 40.80	1.50
6 segments, accepted	21.48	-11.0%	39.13 / 40.79	1.42
Perfect-mixing optimum	16.01			
Energy lower bound	15.41			

```

Input: network, limits, K, starting schedules
for each starting schedule x0:
    minimize sum(x)*segment_time over 0 <= x <= 3 L/min
    subject to min_i T_i(t) >= Tmin+reserve,
               max_{i in Omega} T_i(t) <= Tmax,
               spread over Omega <= span (every 5 s)
retain coarse candidates and buffered alternatives;
replay on three meshes with independent integration;
accept only if continuous bounds meet all limits

```

Figure 3 shows flow timing and temperature constraints.

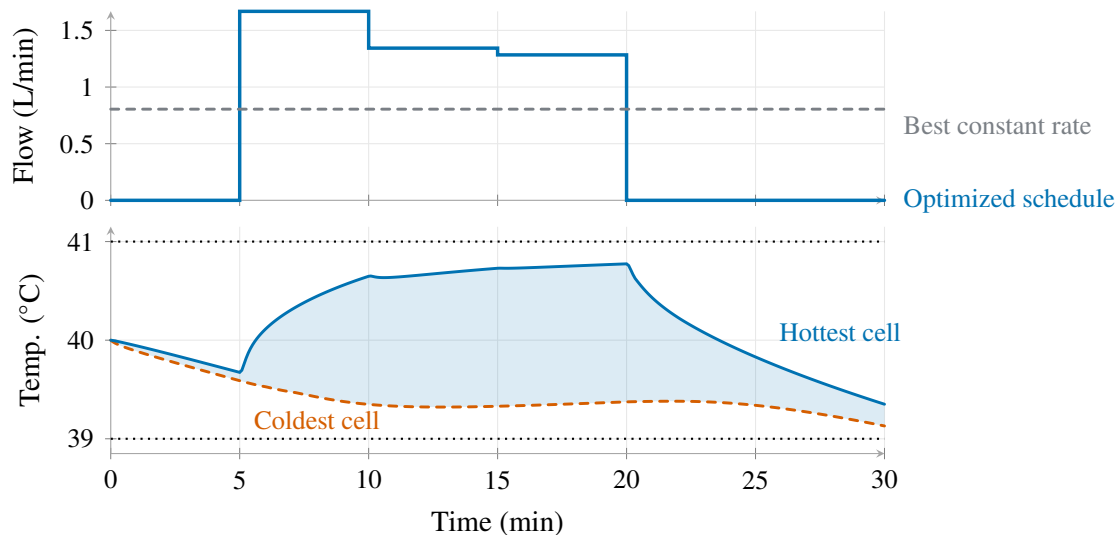


Figure 3. Selected buffered 6-segment schedule against the best constant rate (top) and the range of cell temperatures outside the inlet jet zone (bottom); dotted lines mark the limits. The baseline concentrates inflow in the middle 15 minutes; both no-flow periods enter the independent checks.

The accepted 6-segment policy is off for the first 5.0 minutes and last 10.0 minutes, with a peak flow of 1.67 L/min. For consecutive 5-minute intervals the rates, rounded to nine decimals, are 0, 1.668107192, 1.343440056, 1.283871597, 0, 0 L/min; the interval duration times their sum is

21.477094 L. It adds 21.48 L, 11.0% below the constant rate. Its 0.1°C design margin sacrifices some water savings to protect the stated constraints. The remaining 6.06 L gap to the energy bound is not an optimality certificate.

The unbuffered 12-segment candidate uses 19.35 L but reaches a spread of 1.531°C on the finest mesh, above the 1.5°C limit, so it is rejected. Among the tested buffered alternatives, the accepted 6-segment policy gives spreads of 1.400 , 1.395 and 1.416°C on the $8 \times 4 \times 3$, $12 \times 6 \times 4$ and $16 \times 8 \times 6$ meshes. Independently integrated continuous-time bounds also pass on each mesh. This establishes numerical feasibility for these tested networks, not mesh-independent physics or a global minimum.

7.2 What does staying close to the start temperature cost?

The task asks for a bath close to its initial temperature without wasting much water, which is a trade-off rather than a single optimum. Figure 4 repeats the search while the allowed fall below 40°C changes from 0.25 to 3°C ; the 1°C case is the baseline. With a fall of 2.5°C or more the stored heat suffices and no water is added. At 2°C the best constant rate adds 1.9 L, at 1.5°C 12.9 L, at 1°C 24.1 L and at 0.75°C 29.8 L. Between 2°C and 0.75°C each further 0.5°C of tolerance is therefore worth roughly 11 L, almost linearly. At 0.5°C the best constant rate adds 36.3 L; for a fall of 0.25°C no constant-rate policy was accepted, while the perfectly mixed bath would still need 30.7 L.

Figure 4 compares searched spatial policies with analytical benchmarks.

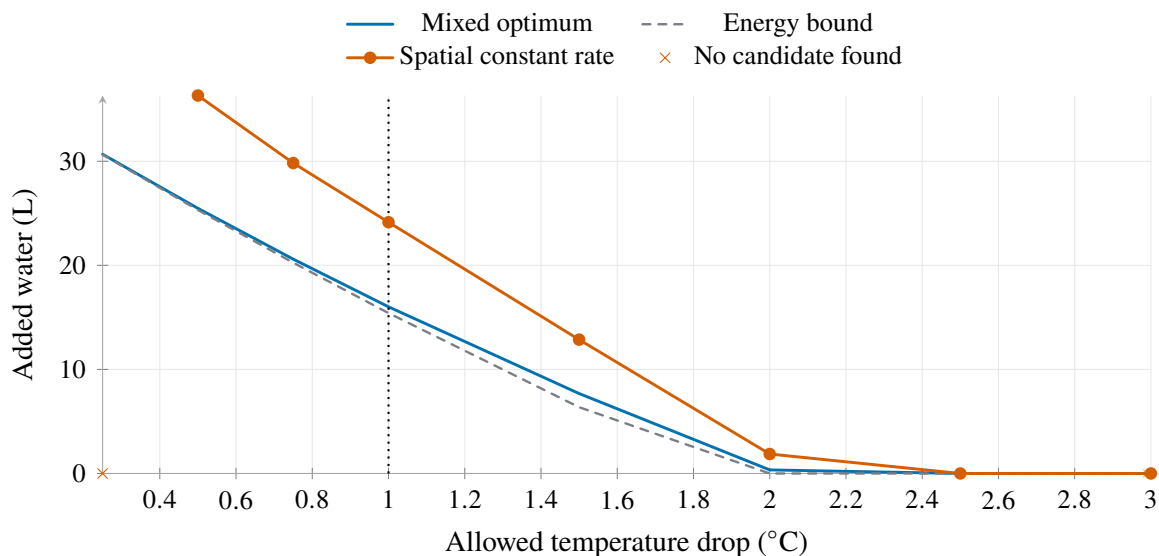


Figure 4. Water use versus temperature tolerance. Crosses denote unsuccessful searches, not zero water demand; the dotted line marks the baseline. Allowing a larger fall sharply reduces demand in the tested constant-rate family.

The spatial-minus-mixed gap increases from 1.5 L at a 2°C fall to 9.2 L at 0.75°C . Scheduling lowers the baseline by 11%; schedules at other tolerances were not optimized.

7.3 Can a user follow the schedule?

We distinguish fixed tap bias from random segment errors. For the unbuffered 12-segment candidate, multiplying every rate by 0.9 or 1.1 crosses a limit by 0.076 and 0.126°C. Independently drawing each multiplier from a normal distribution of mean 1 and standard deviation 0.1, clipped below zero, gives median and 5th-percentile slacks of -0.070 and -0.217°C (200 draws, seed 11). Standard deviation 0.1 does not mean an error bounded within $\pm 10\%$. Negative slack is a violation, however small.

The error is not symmetric. The most sensitive segment is segment 4 (2.81 L/min), where raising the flow by 20% breaks the limits by 0.245°C and lowering it by 20% by 0.047°C: extra hot water overheats the inlet region and widens the spread. Over-delivery in a high-flow segment is the error to guard against. 1 of the 5 active segments tolerates a 20% error in either direction.

The price of a margin is measured on a 6-segment version. Requiring all three limits to hold with a buffer b gives 19.50 L at $b = 0$, 21.48 L (+10%) at 0.1°C, 23.81 L (+22%) at 0.2°C and 26.83 L (+38%) at 0.3°C. With the 0.1°C buffer, 74% of 200 draws with independent Gaussian multiplier standard deviation 0.1 stay within the limits. The 0.1°C policy is selected here for its independently checked numerical margin. Its 74% acceptance under the stated random error model does not make it a reliable manual prescription; a real bath needs temperature feedback and separate calibration.

7.4 Measurement ambiguity changes the decision

Mixing changes timing: the weaker-mixing candidate starts immediately. Two probes miss violations in two of nine cases. We now propagate observation ambiguity into the supply decision.

A separate synthetic trial applies (0, 0, 1.2, 0, 0, 0) L/min in five-minute stages. Known geometry starts uniformly at 40°C; cells 89 and 6 are read every 30 s. Assumed reading errors and constant probe biases are each bounded by $\pm 0.02^\circ\text{C}$; these are not instrument specifications.

For parameters θ and probe j , let $d_j(t; \theta) = T_j(t; \theta) - y_j(t)$. At observed times $\mathcal{T} = \{0, 30, \dots, 1800\}$ s, a model is compatible when a constant correction a_j exists such that

$$\min_{|a_j| \leq 0.02} \max_{t \in \mathcal{T}} |d_j(t; \theta) - a_j| \leq 0.02 \quad (j = 1, 2). \quad (18)$$

The extreme-discrepancy midpoint clipped to the bias bound is the minimax correction, with physical bias $b_j = -a_j$. A trace from -0.04 to +0.04°C fails: bias cannot vary between readings.

Our grid has 7 mixing values (0.00085–0.00115 m²/s), 9 surface and 9 body coefficients (20–30 and 15–35 W/(m² K)), and 5 delivery multipliers (0.95–1.05). Of 2835 models, 178 fit. This finite decision ambiguity set is neither a confidence region nor a probability distribution.

Table 8 compares archived policies with a replacement optimized on ten extreme/failing models, using a 2 L/min bound, 0.1°C target and 0.03°C extra floor reserve. All 178 pass three-grid envelopes; the tightest spread bound is 1.496828°C. The selected target is not an all-model margin or global optimum.

Table 8. Policies on the same measurement-compatible set

Policy	Command (L)	Delivered (L)	Sampled pass
Constant	24.14	22.94–25.35	130/178
Nominal six-stage	21.48	20.40–22.55	174/178
Ambiguity-set six-stage	23.48	22.31–24.66	178/178

The table uses the same 178 models and five-second sampling; archived policies were not re-optimized. Only the replacement has 534 three-grid envelopes. Its five-minute rates are 0.892820, 1.473786, 1.179268, 1.150408, 0.000000, 0.000000 L/min; envelope extrema are 39.1297/40.9238°C.

The replacement costs 9.33% more than the nominal policy. A separate zero-inlet trial with the same prior, probes and errors retains 510 models, including unidentified delivery bias; its candidate passes 1530 three-grid envelopes. The two observation sets intersect in 167 models; they are not nested or probabilistic.

Table 9 separates control and trial water. Both candidates restart at uniform 40°C with the same control family and design targets.

Table 9. Observation-conditioned trial and control costs

Observation / trial	Models	Control (L)	Trial (L)
Nominal / passive	510	24.39	0.00
Nominal / pulse	178	23.48	6.00
Strong mixing / passive	60	16.51	0.00
Strong mixing / pulse	14	15.37	6.00
Weak mixing / passive	70	Not accepted	0.00
Weak mixing / pulse	11	35.05*	6.00

Passive rates are 1.419252, 1.083923, 1.055204, 1.044908, 0.274962, 0.000000 L/min. Initial filling and reset are not free; these conditional candidates do not establish expected information value or global optimality.

Let n unchanged baths share one trial; let $\alpha \in [0.95, 1.05]$ be the common delivery multiplier and R_0, R_1 the actual reset litres for passive and pulse designs. The difference between their total actual-water costs is

$$C_0 - C_1 = \alpha(n\Delta - 6) + R_0 - R_1, \quad \Delta = 0.909836 \text{ L.} \quad (19)$$

At nominal delivery, passive costs less for one bath if $R_0 - R_1 < 5.0902$ L. Equal reset costs give a crossover at 6.59 baths: passive is cheaper through six, pulse from seven. Reuse requires unchanged geometry, heat transfer, probes and delivery bias; otherwise remeasure. This is not online feedback or a real-bath guarantee.

Two additional observations use $(D, \text{ surface/body coefficients})=(0.00115, 20/15)$ and $(0.00085, 30/35)$, in m^2/s and $\text{W}/(\text{m}^2 \text{ K})$, with nominal delivery. Separate trials retain the same prior/errors. Strong mixing gives a 1.1406 L control-cost gap, so equal-reset pulse reuse pays from six. The three accepted candidates pass 255 distinct three-grid envelopes.

*The weak-mixing pulse passes physical limits but fails the optimizer and extra design margin. No passive candidate was accepted within two starts and three constraint-generation rounds; neither failure proves infeasibility. These are observation-conditioned outcomes, not ex ante experiment values or sequential inference.

8 Separate geometry, size and body effects

Table 10 isolates geometry, storage and body changes.

Table 10. Geometry, size and body scenarios

Change from baseline	Water (L)	Interpretation
Shallow Wide	No candidate	Same envelope volume
Deep Narrow	17.48	Same envelope volume
Small Bath	27.05	Lower water depth
Large Bath	20.39	Higher water depth
Deep Bath	13.05	Depth 0.35 m, like a bath filled to the shoulders
Larger Body	28.11	More displacement/contact
Long Body	23.85	Shape only; fixed volume/area
Warmer Skin	21.08	Skin at 36°C
Cooler Skin	27.18	Skin at 32°C

At equal volume, the wide, shallow tub has a larger exposed top area and a changed diffusion length; it has no accepted candidate. The deep, narrow tub needs 17.48 L against 24.14 L, because a deep, narrow envelope reduces top-area heat loss, although a real deep bath may stratify, which Section 9 tests with a separate vertical-mixing scenario. Geometry is therefore more than an interchangeable cooling coefficient.

Changing depth changes storage and shell area. The large bath needs 20.39 L replenishment; the small bath needs 27.05 L. At 0.23 m the baseline is shallow: its surface-to-volume ratio is about 1.7 times that at shoulder depth; a 0.35 m bath needs 13.05 L. Initial filling is excluded from J, so lower replenishment does not establish lower total water use.

The larger-body case combines displacement and contact-area changes and needs 28.11 L; the long-body case changes distribution at fixed volume/area. Warmer skin reduces the exchange driving difference. These are conditional scenarios, not measured human responses.

To transfer the model to a real tub, measure water volume after entry, submerged contact geometry, temperatures at several depths and distances, and no-inlet cooling. Shape and temperature should not be inferred from a single mean cooling curve.

9 Motion and a bubble layer can change the policy

Table 11 separates transport benefits from boundary losses.

Table 11. Mixing, surface, comfort and supply scenarios

Scenario	Added water (L)	Search outcome
Weak Mixing	—	None accepted
Strong Mixing	19.40	Accepted
Moving With Added Surface Loss	23.04	Accepted
Stratified	24.42	Accepted
Convective Surface Layer	24.10	Accepted
Foam	10.37	Accepted
Tight Comfort	36.32	Accepted
Loose Comfort	1.87	Accepted
Low Loss	8.60	Accepted
High Loss	—	None accepted
Cool Supply	44.80	Accepted

Increasing D reduces the gradient created by the localized inlet. The strong-mixing scenario uses 19.40 L against 24.14 L, but motion may also increase heat loss. When D is increased together with a 20% increase in the surface coefficient (30 instead of 25 W/(m² K)), the need rises to 23.04 L and most of the benefit disappears. The comparison deliberately separates transport improvement from its possible boundary cost.

Stratification is tested separately. A hot inlet layer is lighter than the water below, and the stable density gradient suppresses vertical mixing. With a cell height of 0.077 m, a vertical temperature difference of 1 K, a velocity scale of 0.02 m/s and an expansion coefficient of about 3.8×10^{-4} per K, the gradient Richardson number is $Ri = g\beta \Delta T \ell / u^2 \approx 0.7$. The Munk–Anderson stability function for scalars, $(1 + 3.33 Ri)^{-3/2}$ [8], then reduces vertical diffusivity to about 0.15 of its neutral value. We test a vertical-to-horizontal ratio of 0.2. The best constant rate becomes 24.42 L against 24.14 L, but the maximum spread rises from 0.97 to 1.07°C against the 1.50°C limit. Stratification therefore costs little water here and uses 19% of the remaining uniformity margin; the ratio is an order-of-magnitude scenario and the Richardson estimate uses assumed velocity and temperature scales. The stable layer is an upper-bound picture: evaporative cooling at the surface can instead drive convection and raise vertical mixing, so a vertical-to-horizontal ratio of 3 is also tested, and it needs 24.10 L.

The foam scenario multiplies only the effective surface coefficient by 0.4. With all other inputs fixed, the best constant rate falls to 10.37 L. The assumed 60% reduction is a scenario, not an experimentally established property of bubble-bath additive. If the layer breaks up or motion raises evaporation, this result must be recomputed.

No candidate was accepted for: Weak Mixing, Shallow Wide, High Loss. In weak-mixing, high-loss and wide-shallow trials, sufficient inflow violates spread on sampled rates 0.8–2.5 L/min.

Finite search does not prove infeasibility. Circulation, inlet layout, duration or comfort tolerance may need to change; each remedy requires its own evidence.

10 Validation: independent evidence and remaining error

All 17 recorded checks pass. Independent heat-flow arithmetic sums environmental losses, body exchange and local-temperature overflow; the largest instantaneous residual is 2.46×10^{-10} W. An adaptive RK45 integration with independently assembled right-hand side agrees with the archived matrix-exponential trajectory within $3.12 \times 10^{-07}^\circ\text{C}$. This tests numerical implementation, not a second physical model.

Uniform-loss analytic cooling and a single conservative overflow sink check signs and energy; nonnegative off-diagonals and nonpositive row sums give contraction within each flow segment.

Half-second replay is supplemented by a derivative bound: if $|dT_i/dt| \leq L$, a point lies at most 0.25 seconds from a sample, so temperatures differ by at most $L/4$ and spread by at most $L/2$. With an integration allowance and no relaxation of the physical limits, the resulting lower envelope is 39.0144°C , upper 40.3340°C , and spread upper bound 0.9980°C . This is a conditional floating-point envelope, not interval-arithmetic certification.

The accepted schedule uses an independently assembled right-hand side and adaptive RK45, restarted at every flow switch. Nonnegative off-diagonal transport and nonpositive row sums give an infinity-norm contraction bound for the derivative within each segment. One-second samples alone leave some intervals unresolved; adding RK45 step endpoints and bisecting unresolved intervals once on the finest mesh gives bounds of 39.1297°C for the minimum, 40.8304°C for the maximum and 1.4964°C for the spread. These pass the physical limits without threshold relaxation. A $2 \times 10^{-6}^\circ\text{C}$ integration allowance is included; this floating-point check is conditional on the network, not interval arithmetic or experimental validation. The 17 baseline checks are separate.

11 Resolution, uncertainty and transfer to a real bath

Table 12 compares constrained temperatures and the excluded inlet peak.

Table 12. Mesh replay: constrained temperatures

Mesh (cells)	Max outside jet zone	Spread outside zone	Inlet-cell max
$8 \times 4 \times 3$ (96)	40.318°C	0.967°C	40.553°C
$12 \times 6 \times 4$ (288)	40.315°C	0.960°C	40.757°C
$16 \times 8 \times 6$ (768)	40.330°C	0.973°C	40.995°C

The same constant-rate policy is replayed on three meshes. The ceiling and spread quantities outside the jet zone agree within 0.015°C and 0.013°C , and the tested constant-rate trajectory stays within the comfort limits on these meshes. The inlet-cell maximum rises from 40.55 to 41.00°C .

with refinement; that is why the cell is excluded rather than constrained. The selected water amounts on the $8 \times 4 \times 3$ and $12 \times 6 \times 4$ meshes are 24.142 and 24.117 L (0.11% apart). Three meshes are a diagnostic, not an asymptotic convergence-order study.

Parameter uncertainty concerns heat loss, body exchange and mixing. Structural uncertainty concerns the prescribed surface stream, fixed skin reservoir, homogenized body and constant effective D . These are different errors. The scenario table explores parameter dependence and one coupled motion effect; it is not a probability distribution or confidence interval.

Among 64 scrambled Sobol draws over the six stated input ranges (D log-uniform), 41 admit a searched constant-rate candidate. Water quantiles at 5/25/50/75/95% are 12.5/17.0/22.0/26.9/30.9 L. These scenario quantiles exclude 23 failed searches and are not real-bath probabilities.

Spearman rank correlations with water, over accepted draws: surface coefficient +0.63, room temperature -0.47, body coefficient +0.47, skin temperature -0.27, shell coefficient +0.15, mixing diffusivity -0.15.

Accepted/rejected draws have median D $1.4 \times 10^{-03}/4.5 \times 10^{-04}$ m²/s (Mann–Whitney $p < 1 \times 10^{-4}$); the other five inputs show no detected association (smallest $p=0.47$). Weak mixing can leave distant water cold while extra inflow violates uniformity.

An inlet-free trial cheaply characterizes cooling but cannot identify a term it never excites. At uniform initial T_0 , internal exchange vanishes, so $\dot{T}_i(0) = [H_{a,i}(T_a - T_0) + H_{b,i}(T_b - T_0)]/C_i$, independent of D . Later spatial differences can reveal mixing. Delivery bias and inlet path remain absent throughout the passive trial. A warming contact reservoir shares the initial slope but changes later curvature; loss parameters can compensate for that difference. An inlet trial must therefore check transport and overflow energy, not merely refine a cooling fit. Reserve a separate experiment to check predictions; match motion/foam trials for changes in loss as well as mixing.

The finite-volume stencil is also checked against an insulated three-dimensional cosine diffusion mode: four successively refined grids approach second order (finest observed order 1.98). A zero-diffusion inlet path matches the independent stirred-cell cascade step response within $4.3 \times 10^{-14}^\circ\text{C}$. These verify implementation in empty-domain limits, not the mixing closure or real baths [9].

11.1 Carry structural ambiguity into the decision

The alternatives replace fixed skin by $C_b \dot{B} = \sum_i H_{b,i}(T_i - B)$ (253.31 kJ/K, initially 34°C), or route the inlet stream down, across the bottom and back to the same overflow. Contact storage is not physiology; the fixed-skin energy bound does not transfer.

Eight alternative-mechanism observation sets fit the original structure; six existing policies pass specified sampled replays, with min/max/spread/final-body summary differences below $1.85 \times 10^{-08}^\circ\text{C}$. We therefore screen surface/fixed, deep/fixed and surface/finite hypotheses together: passive/pulse banks retain 1465/467 pairs. Each trial restarts uniformly; observations are not sequential.

Base-grid success hid two finest-grid ceiling failures (41.0010/41.0007°C). A 1% passive-flow reduction repairs those sampled cases, without common reserves or full fine-grid coverage. New

six-stage candidates instead target 39.13°C floor, 40.9°C outside-zone ceiling and 1.4°C spread across all three grids. These extra reserves leave the physical 39/41/1.5°C limits unchanged.

Table 13 compares structure-specific banks at the same reserves.

Table 13. Common-reserve structural decisions

Structure / trial	Pairs	Control (L)	Floor / ceiling (°C)	Spread (°C)
Surface/fixed / passive	510	26.09	39.132 / 40.857	1.374
Surface/fixed / pulse	178	23.63	39.132 / 40.900	1.400
Deep/fixed / passive	510	26.09	39.132 / 40.900	1.400
Deep/fixed / pulse	149	23.63	39.133 / 40.900	1.400
Surface/storage / passive	445	26.09	39.141 / 40.848	1.374
Surface/storage / pulse	140	23.63	39.145 / 40.895	1.400

The local search uses exact affine sensitivities on active cells. Two base-grid members fail the common floor despite passing physical limits. Adding those constraints gives the candidates in Table 13. For consecutive five-minute stages their rates are 1.303664, 1.053753, 1.011697, 0.988372, 0.860138, 0.000000 L/min (passive), and 1.126406, 1.318679, 1.149624, 1.130662, 0.000000, 0.000000 L/min (pulse). Unrounded settings are retained for computation.

All 5796 model-grid objects pass conditional continuous-time envelopes. The dynamic temperatures $U = (T_1, \dots, T_N, B)$ exclude the constant coordinate. Nonnegative off-diagonals and nonpositive row sums give contraction within each flow stage. Chord error is bounded by $\|U''(t_0)\|_\infty h^2/8$; the first derivative gives another enclosure. Unresolved one-second intervals are bisected to at most 0.0625 s. A 2×10^{-6} °C allowance (twice for spread) is assumed, not interval-proved. Table 13 rounds outward. Independent RK45/energy checks cover 54 extremal identities, not the entire bank.

Section 7.4's cost identity now has $\Delta = 2.461267$ L, giving $n > 2.43777$: pulse beats the fixed passive candidate from use 3, if one 6 L trial is reused, delivery bias stays positive and fixed, and total reset costs match. The seven-use result was conditional on the original structure. This expanded-bank comparison establishes neither global optimality nor expected experiment value.

11.2 Use observations to change the next action

Starting from 1,465 passive-compatible pairs on the 96-cell grid, read probes 89 and 6 every minute. Given prediction $\widehat{T}_{mj,k}$ and reading $z_{j,k}$, retain each model's constant-bias intervals:

$$I_{mj,k} = [-0.02, 0.02] \cap \bigcap_{\ell=0}^k [z_{j,\ell} - \widehat{T}_{mj,\ell} - 0.020002, z_{j,\ell} - \widehat{T}_{mj,\ell} + 0.020002]. \quad (20)$$

Exclude a model if either interval is empty. Choose the least of 0, 0.8, 0.9 and 1 times the current backup rate whose next minute and remaining backup satisfy all survivors' common targets. Chord bounds include the dynamic body state, refine unresolved five-second intervals to 0.0390625 s, and

assume the same $2 \times 10^{-6}^\circ\text{C}$ allowance.

Conditional feasibility and water cap. Each accepted minute is checked with the original remaining backup. Subsequent observations only shrink the set, preserving that feasible tail. Induction applies if the truth stays in the static bank and propagators are valid. For L/min commands $0 \leq q_k \leq q_k^{\text{backup}}$, a completed service has $J_{\text{cmd}} = \sum_k q_k \Delta t \leq \sum_k q_k^{\text{backup}} \Delta t \leq 26.09 \text{ L}$ ($\Delta t = 1$ minute). A shared $\alpha > 0$ gives $J_{\text{act}} = \alpha J_{\text{cmd}} \leq \alpha \times 26.09 \text{ L}$. This proves neither minimum water nor fault or timeout recovery.

Fixed probe biases $(+0.013, -0.011)^\circ\text{C}$ give 23.59 L with zero reading noise and 23.16 L with one bounded-noise sequence (seed 20261010); 114/1 models remain. Both start from 1,465 hypotheses and the 26.09 L backup. Figure 5 shows action changes before unique identification. These are two nominal surface/fixed examples, not population performance.

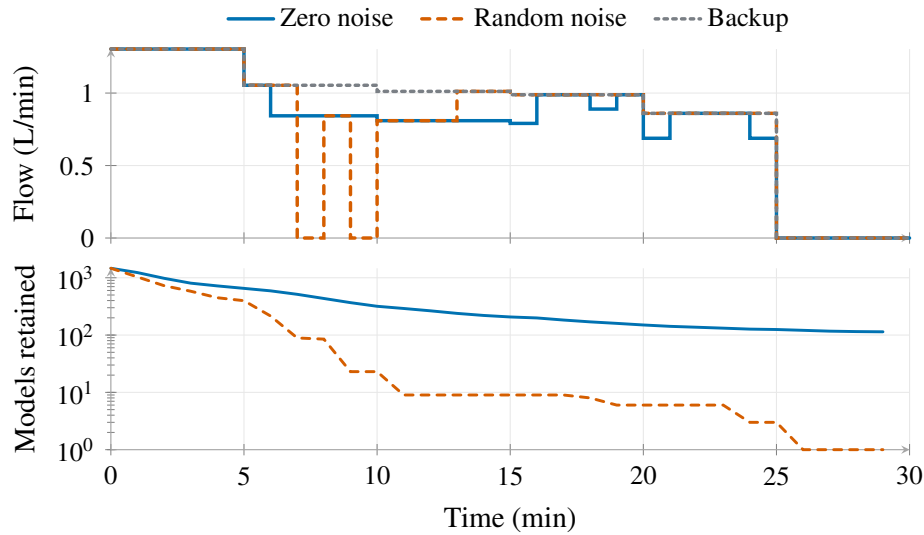


Figure 5. Observations change replenishment. Both nominal cases start from the same bank and backup; a logarithmic count axis distinguishes slow and rapid elimination. The constant bias and noise bounds are shared.

A separate matched 12-model ablation uses 26.00 L without observations versus 22.86/22.86/22.08 L with them in three structures. Six feedback trajectories pass 18 independent half-second three-grid RHS/energy replays; realized actions do not certify fine-grid feedback throughout the bank.

Minute-10 shifts to 45°C supply or $40 \text{ W}/(\text{m}^2 \text{ K})$ surface loss empty the set at minute 11; terminal bias intervals independently exclude all 23 preceding survivors. Neither partial service counts as savings. An empty set invalidates reuse, without identifying the cause or certifying recovery.

11.3 Check transfer beyond the parameter grid

Six predeclared bank-external truths give two complete services (22.59/22.80 L commands). Changed storage, dimensions, supply or loss empties the bank after 28/5/11/9 minutes, before

a checked temperature violation. Partial services are not savings; refusal supplies no recovery policy.

For the two completed surface/deep paths, fix the realized commands, baseline geometry, uniform 40°C water, 34°C fixed contact, 50°C inlet and 22°C air. Around $(D, h_s, h_b, \alpha) = (0.001025, 24.375, 26.25, 1.0125)$, the respective halfwidths are 6.25×10^{-06} m²/s, 0.15625 and 0.3125 W/(m² K), and 0.003125. Parameters are static; these are mathematical neighborhoods, not measured tolerances.

Fixed-command perturbation bound. Write $U' = M_\theta U + c_\theta$, affine in the four parameters. For halfwidths w_j , set $K_j = w_j \partial_j M$, $f_j = w_j \partial_j c$, and $\theta = \theta_0 + w \odot z$ with $|z_j| \leq 1$. Nominal variations satisfy $S'_j = M_0 S_j + K_j U_0 + f_j$, $S_j(0) = 0$.

Proof. The exact remainder $r = U_\theta - U_0 - \sum_j z_j S_j$ satisfies

$$r' = M_\theta r + \sum_{j,k} z_j z_k K_j S_k, \quad r(0) = 0. \quad (21)$$

Positive parameters give nonnegative off-diagonals and nonpositive row sums at every vertex, hence throughout the affine box. Contraction [10] and variation of constants give

$$\|r(t)\|_\infty \leq R(t), \quad R(t) = \sum_{j,k} \int_0^t \|K_j S_k(u)\|_\infty du. \quad (22)$$

The cellwise radius is $\sum_j |S_j| + R$: subtract it for the floor, add it for the ceiling, and twice for spread. States are continuous at flow switches; R accumulates across them.

□

Half-second derivative/chord bounds enclose the intervals; assumed errors are 2×10^{-6} °C, 2×10^{-6} °C/s and 2×10^{-6} °C/s² for states and first/second derivatives. The tested halfwidth scales were 4, 2, 1 and 0.4 times those above; the displayed box is the largest qualifying both paths on all three meshes: floor ≥ 39.0759 °C, ceiling ≤ 40.7769 °C, spread ≤ 1.1746 °C. Common extra reserves do not all pass. Independent RHS/energy checks cover all 102 corners/centers; finite points challenge implementation, not prove coverage. A prior max-norm defect bound certified none. This exact-remainder result assumes numerical accuracy and certifies fixed actions, not all feedback branches or real baths.

11.4 Challenge the transport mechanism

A separate whole-section axial closure matches baseline volume, storage, temperatures and total conductances. Uniform/localized-body loss profiles test distribution at fixed totals; local capacity, anatomy, vertical structure and jet domains differ from the network. With $v=qL/V$, loss per length λ and capacity per length c ,

$$T_t = DT_{xx} - vT_x - \lambda(x, T)/c, \quad vT(0) - DT_x(0) = vT_{in}, \quad T_x(L) = 0. \quad (23)$$

The flux inlet [11] and zero-gradient outlet conserve enthalpy. Three meshes (40/80/160 cells) replay both original policies for each profile and $D=0.0003/0.001/0.003$ m²/s. Weak mixing defeats both policies: with constant inflow the mean ends at 39.444°C but the outlet at 38.581°C. Medium/strong cases pass sampled physical checks. Table 14 gives a separate N80 first-crossing delayed-constant search; unchanged candidates pass 1 s replays on all three meshes.

Table 14. Axial-closure candidates

D (m ² /s)	Start (min)	Water, uniform / body-local (L)
0.001	0	22.71 / 22.68
0.003	10	18.87 / 18.86

Starts are 0:120:1200 s, first floor roots are bracketed on 0:0.2:3 L/min, with design floor 39.03°C. At $D=0.0003$ neither profile has an accepted root; this is finite search failure. The result supports timing dependence, not a universal start or global optimum.

$$E_{\text{excess}} = \rho c_p \int q(T_{\text{out}} - \bar{T}) dt. \quad (24)$$

Positive excess means greater overflow loss than at the contemporaneous mean. The original constant-rate baseline gives -31.51 kJ; the matched axial uniform case gives -26.28 kJ. Both outlets are cooler than the mean. A permitted short circuit is therefore not evidence that it caused the baseline water penalty.

For uniform loss, $k = (H_a + H_b)/C$ and $T \leq T_{\text{in}}$. For $w = T_x$, $w_t = Dw_{xx} - vw_x - kw$, $w(L) = 0$ and $w(0) = v[T(0) - T_{\text{in}}]/D \leq 0$. The maximum principle, uniform initial water and inherited interior gradient at each switch give $w \leq 0$, hence $T_{\text{out}} \leq \bar{T}$. Localized loss adds gradient forcing. A cold outlet can retain heat while a distant region violates comfort.

Steady ODE/Robin eigenfunction answers show first-order FV convergence (N160 transient error $\leq 0.002614^\circ\text{C}$); six independent flux/RK45 traces differ $\leq 3.30 \times 10^{-07}^\circ\text{C}$, energy ≤ 0.00213 J. This is reduced-model verification, not physical or continuous-PDE certification. Diagnose outlet enthalpy before prescribing a short-circuit remedy.

12 Conclusions and a policy with clear scope

Choose tolerance, then check transport. Figure 4 prices warmth; both closures show remote cooling and timing dependence without positive excess overflow loss in their baselines. Geometry/body contact change demand; motion can raise loss, foam reduces it.

Baseline/scheduled replenishment is 24.14/21.48 L. Common-reserve passive/pulse candidates use 26.09/23.63 L; reusable 6 L calibration pays from use 3 at equal resets/unchanged conditions. Feedback yields 23.59/23.16 L against a 26.09 L cap in two nominal cases. Fixed-action bounds and refusals do not certify full feedback, recovery or hot-jet safety.

13 A warmer bath, with less replacement water

A guide for the person in the bathtub

Choose your tolerance. Allowing some cooling uses stored heat before replacement water. Our example needs about 24 L with a 1°C allowance, but 1.9 L with 2°C. These are model scenarios; your comfort and health needs determine the acceptable range.

Redistribute warmth. Water near the faucet may warm while distant water stays cool. Gentle movement can help; stronger motion can increase surface loss. Extra inflow can widen temperature differences. A cooler overflow can retain heat while a distant cold patch persists; measure both before attributing poor performance to hot-water short circuit.

Use timing only under its stated conditions. The baseline schedule waits 5 minutes, supplies water for 15 minutes and stops for 10 minutes. It uses 21.48 L against 24.14 L for a constant trickle. Weaker mixing can require inflow immediately. Even with a margin, only 74% of the specified random tap-error trials stayed within limits; this is not a reliable manual prescription.

Let readings change the plan. Our modeled controller checks two temperatures every minute, then reduces inflow only if the entire remaining plan passes for every compatible model. A unique mechanism need not be identified first. If no model explains the readings, stop relying on that certificate and reassess supply, motion and heat loss. Two specified shifts were detected after one minute; this is not a universal alarm or a tested emergency response. Two probes can also miss a cold corner outside the modeled conditions.

Count preparation as well as use. Across the broader model bank, cooling-only and hot-water trials lead to 26.09/23.63 L during a bath. For those fixed schedules, a reusable 6 L hot-water trial pays back from use 3 when conditions and reset costs match. Both trials need 30 minutes and a fresh uniform start. The online cases reduce the passive backup to 23.59/23.16 L under specified sensor errors; their repeated-use crossover with pulse is untested; trial/reset costs remain.

Keep the practical limits in view. An intact bubble layer may reduce heat loss, but the calculation does not measure a particular additive. Changed geometry, body position, motion, supply or foam needs reassessment. Use stored warmth, keep it distributed and avoid unnecessary overflow; the figures explain these tradeoffs rather than certifying bathing or hot-jet safety.

14 References and reproducible algorithm

- [1] COMAP. 2016 MCM Problem A: A Hot Bath. 2016. Official problem PDF: contest.comap.com/undergraduate/contests/mcm/contests/2016/problems/2016_MCM_Problem_A.pdf.
- [2] U.S. Department of Energy. DOE Fundamentals Handbook: Thermodynamics, Heat Transfer, and Fluid Flow. DOE-HDBK-1012/2-92, June 1992. Convection heat transfer, Eq. (2-9).
- [3] IAPWS. Revised Release on the IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use. R6-95(2018). iapws.org/technical-guidance/release/IAPWS-95.
- [4] SciPy developers. `scipy.sparse.linalg.expm_multiply` and `scipy.integrate.solve_ivp`, API documentation. docs.scipy.org/doc/scipy/reference/. Accessed 7 October 2026.
- [5] Bergman, T. L., Lavine, A. S., Incropera, F. P., DeWitt, D. P. Fundamentals of Heat and Mass Transfer, 7th ed. Wiley, 2011. Natural convection above a heated horizontal surface (characteristic length A/P); heat and mass transfer analogy.
- [6] Menzies, C., Clarke, N., Steward, C. J., Thake, C. D., Pugh, C. J. A., Cullen, T. Vascular, inflammatory and perceptual responses to hot water immersion: impacts of water depth and temperature in young healthy adults. *Experimental Physiology*, 2025. doi:10.1113/EP092761.
- [7] OpenAI, Codex (GPT-6-based assistant), and Anthropic, Claude Sonnet 5.5 in Claude Code. Used during 7–10 October 2026 for modeling, code, validation and report composition; exact builds not independently established. See Report on Use of AI.
- [8] Munk, W. H., Anderson, E. R. Notes on a theory of the thermocline. *Journal of Marine Research*, 7(3), 276–295, 1948.
- [9] NASA/NPARC Alliance. Verification Assessment and Validation Assessment, CFD tutorial. www.grc.nasa.gov/WWW/wind/valid/tutorial/. Accessed 9 October 2026.
- [10] Higham, N. J. What Is the Logarithmic Norm? 18 January 2022. nhigham.com/2022/01/18/what-is-the-logarithmic-norm/. Theorems 3 and 6. Accessed 10 October 2026.
- [11] COMSOL. Theory for the Inflow Boundary Condition. Heat Transfer Module documentation, version 6.3. doc.comsol.com/6.3/doc/com.comsol.help.heat/heat_ug_theory.07.009.html. Accessed 10 October 2026.

Algorithm and computational record

Reject excessive cell occupancy; assemble capacities, symmetric exchange and a conservative stream. Compute mixed benchmarks; bracket delayed-constant floor crossings, reject ceiling/spread violations, and rank accepted water costs. Optimize buffered schedules and qualify them on three meshes. Recompute searches at Sobol draws. Independent RHS/RK45 uses $\text{rtol } 2 \times 10^{-9}$, $\text{atol } 2 \times 10^{-10}^\circ\text{C}$ and steps ≤ 5 s, with energy/analytic checks. Section 11.4 separately tests an axial closure. AI use is recorded in [7].

Appendix A. Scenario definitions

Each row changes only the listed baseline inputs. All unspecified inputs remain as stated in Section 2; geometry and conductance are then rebuilt. The equal-volume aspect-ratio cases retain the envelope volume of 0.22425 m³. Widths s describe a smooth displacement distribution, not measured anatomical semiaxes.

Accepted candidates are independently replayed on their scenario network at intervals of no more than five seconds. These scenario checks are sampled checks; continuous-time envelopes are reported for the baseline constant rate and the selected scheduled policy; the latter uses additional interval refinement when necessary. “None accepted” refers to the documented finite search and is not an infeasibility theorem.

Table 15 lists the changes defining each scenario.

Table 15. Scenario definitions

Scenario	Changed input(s)
Weak Mixing	D (m ² /s) = 0.0003
Strong Mixing	D (m ² /s) = 0.003
Moving With Added Surface Loss	D (m ² /s) = 0.003; surface coefficient (W/m ² K) = 30
Loose Comfort	lower limit (°C) = 38
Tight Comfort	lower limit (°C) = 39.5
Foam	surface multiplier = 0.4
Shallow Wide	length (m) = 1.7; width (m) = 0.75; depth (m) = 0.175882
Deep Narrow	length (m) = 1.3; width (m) = 0.6; depth (m) = 0.2875
Small Bath	depth (m) = 0.2
Large Bath	depth (m) = 0.27
Deep Bath	depth (m) = 0.35
Larger Body	body displacement (m ³) = 0.075; body contact area (m ²) = 1.35
Long Body	body widths s (m) = [0.48, 0.12, 0.1]
Warmer Skin	skin temperature (°C) = 36
Cooler Skin	skin temperature (°C) = 32
High Loss	surface coefficient (W/m ² K) = 36; wall coefficient (W/m ² K) = 8.5; body coefficient (W/m ² K) = 40
Low Loss	surface coefficient (W/m ² K) = 17; wall coefficient (W/m ² K) = 4.5; body coefficient (W/m ² K) = 12
Stratified	vertical/horizontal D ratio = 0.2
Convective Surface Layer	vertical/horizontal D ratio = 3
Cool Supply	inlet temperature (°C) = 45

Report on Use of AI

Tool and scope

Codex (GPT-6-based) and Claude Sonnet 5.5 via Claude Code assisted modeling, calculation, validation, figures and writing on 7–10 October 2026; exact builds remain unverified. Post-sealing learning used full text of Outstanding papers 44845/54164, page overviews, selected figures and COMAP commentary; their data/figures were not copied. AI sub-agents supplied independent RHS/probe checks. This was not an unseen-problem trial.

Task

The task was to produce a complete historical MCM case; no conversation wording is reproduced.

Outputs, corrections and verification

Independent integration rejected the 19.35 L schedule and accepted a buffered alternative; fixed-policy replays then checked contact storage and flow paths. Baseline optimization was reused; coefficient/inlet-zone corrections remain recorded. Sources [1]–[4] and [6] were checked directly, and [10] at Theorems 3/6; [5]/[8] were not opened, so their relations were checked indirectly. AI produced the derivations and report from actual numerical results.

Control studies compared known-mixing schedules, pulse/passive ambiguity sets, alternative structures and failed candidates. New common-reserve candidates covered 5796 model-grid objects, with 54 independent extremal replays; two reserve failures were retained and corrected. Archived producers retain their versions. Analytical diffusion and inlet-cascade tests verify discretization. Baseline optimization was reused, and all observations are synthetic.

Finite-bank feedback gave nine complete services/18 replays; two shifts stopped at 660 s. Six off-bank truths gave two services/four refusals and ten envelope replays. AI checked stored empty intervals, not missing predictions. Exact-remainder bounds qualified narrow fixed-action boxes with 102 independent replays; the failed loose bound remains. No human verification or feedback-wide certificate is claimed.

AI derived and tested the axial closure with continuum solutions, independent flux integration and unchanged controls; outlet diagnostics revised the short-circuit interpretation. Source [11] was read; COMSOL was not run. This is a counterfactual, not an experiment.

Record limitations

The artifact retains the task description, tool identity, scope, code and numerical receipts, but lacks a full exported transcript and every intermediate AI output. These records do not substitute for human verification or physical bath experiments.